



Developing Applications with Python - Exam Paper

Duration: 60 minutes | Marks: 15

Question:

The medical service system used to manage patients and set appointments. Any patient could book an appointment. Please use Python language and MySQL database to write the program with functions below:

1. Connection to database.
2. Write some functions to add 3 patients, 5 doctors from the keyboard.
3. Write the function to add 3 appointments for 3 patients.
4. Write a function to make a report with the template below and show it to the screen.

No	Patient name	Birthday	Gender	Address	Doctor name	Reason	Date
1	Nguyen A	2010	Male	Ha noi	Nguyen Si		
2	Nguyen B	1990	Female	Ha noi	Tran Binh		

5. Get all appointments today and show them to the screen.

Address	No	Patient name	Birthday	Gender	Doctor name	Status	Note
Ha noi	1	Nguyen A	2010	Male	Nguyen Si	Pending	
Ha noi	2	Nguyen B	1990	Female	Tran Binh	Done	

Database information:

```
create database medical_service;
```

```
use medical_service;
```

```
CREATE TABLE patients (
```

```
    patient_id INT AUTO_INCREMENT PRIMARY KEY,
```

```
    full_name VARCHAR(255) NOT NULL,
```

```
    date_of_birth DATE NOT NULL,
```

```
    gender VARCHAR(10) NOT NULL,
```

```
    address VARCHAR(255),
```

```
    phone_number VARCHAR(15),
```

```
    email VARCHAR(100)
```

```
);
```

CREATE TABLE doctors (

doctor_id INT AUTO_INCREMENT PRIMARY KEY,
full_name VARCHAR(255) NOT NULL,
specialization VARCHAR(100) NOT NULL,
phone_number VARCHAR(15),
email VARCHAR(100),
years_of_experience INT

);

CREATE TABLE appointments (

appointment_id INT AUTO_INCREMENT PRIMARY KEY,
patient_id INT NOT NULL,
doctor_id INT NOT NULL,
appointment_date DATETIME NOT NULL,
reason VARCHAR(255),
status VARCHAR(50) DEFAULT 'pending',
CONSTRAINT fk_patient_id FOREIGN KEY (**patient_id**) REFERENCES

patients(**patient_id**),

CONSTRAINT fk_doctor_id FOREIGN KEY (**doctor_id**) REFERENCES

doctors(**doctor_id**)

);

Marking Schema:

Question	Mark
Question 1 Connection to database	1
Question 2 Insert patients	3
Question 2 Insert doctors	3
Question 3 Insert appointments	2
Question 4 Make a report	3
Question 5 Get all appointments	2
Bonus Good Coding convention	1
Total	15